

Assignment - I

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Subject - Operating System

Subject code-

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1) Note on ChatGpt.?

• ChatGpt is an AI conversational system created by OpenAI using the generative pre-trained, Transformer (GPT) architecture. It can understand and produce natural language, hold context-aware conversations, and help with tasks such as writing, coding, research, and creative projects.

Core-Tech:- Transformer model with attention mechanisms.

Pre-training on vast text data + RLHF (Reinforcement Learning with Human Feedback) for safer, more helpful replies.

Benefits:- 24/7 availability, fast and scalable, consistent tone, accessible via web/app/API.

Limitations:- Knowledge cutoff without browsing, occasional inaccuracies, privacy concerns if sensitive data is shared.

Applications:- Education, business support, software development, creative projects, research assistance.

2) Note on Gemini.

→ Gemini is Google's family of advanced AI models for language and multimodal tasks (text, images, etc).

• Purpose & Features:- Like other LLMs, it handles conversation, summarization, code, reasoning, and image + text tasks when multimodal. Designed for integration with apps & search.

Architecture & Strengths:- Large transformer-based model optimized for reasoning & multimodal inputs. Tends to integrate tightly with google services for up to date info when enabled.

Use-Cases:- Virtual assistants, writing help, multimodal understanding, research assistance.

Limitations & Cautions:- Same general LLM issues - verify outputs, check for hallucinations, respect data privacy before sending sensitive info.

3/ Note on Unix, Linux, Ubuntu and DOS

→ 1. Unix

- Defination: A powerful, multiuser, multitasking operating system originally developed in the early 1970s at AT&T Bell Labs by 'Ken Thompson', 'Dennis Ritchie', and others.
- Core Idea: Simplicity and probability - "Everything is a file" philosophy
- Key Features
 - Multiuser and Multitasking.
 - Hierarchical file system.
 - Portable, written largely in the C language.
 - Strong security and permission system.
 - Rich set of small command-line tools that can be combined.
- Influence: Basis for many modern OS families: Linux, BSD, macOS, Solaris

2. Linux

- Defination: A Unix Like, open-source operating system Kernel created by Linus Torvalds in 1991
- Nature: Technically just the kernel, but commonly refers to the full GNU/Linux operating system.
- Key Features
 - Free and open source under the GNU General public license (GPL).
 - Multiuser, multitasking, highly secure and stable.

- Supports a wide range of hardware servers, desktops, embedded systems.
- Modular design: Kernel + userland tools (often from the GNU project).
- Distributions ("distros") package the kernel with software (examples: Ubuntu, Debian, Fedora, Arch).

Ubuntu:

- Definition: A Linux distribution (or "distro") based on Debian, maintained by Canonical Ltd., first released in 2004.
- Purpose: User-friendly Linux for desktops, servers, and cloud computing.
- Key Feature
 - Free and open source; regular releases with long-term supports (LTS) versions every two years.
 - Pre-installed graphical user interface (GNOME by default).
 - Strong community support and extensive software repositories.
 - Ideal for beginners as well as professionals (desktop, server, IoT, cloud).
 - Variants like Kubuntu, Xubuntu, Ubuntu Server, Ubuntu Studio.

4 DOS (Disk Operating System):

- Definition: An early single-user, single-tasking operating system primarily for IBM-compatible PCs in the 1980s and early 1990s.
- Examples: MS-DOS (Microsoft), PC-DOS (IBM), Free DOS.
- Key Features -
 - Command-line interface (no built-in GUI)
 - 16-bit architecture, runs on x86 hardware
 - File system support like FAT 16
 - Limited multitasking; each program ran alone
- Historical Role: Foundation for early versions of Microsoft Windows (Windows 3.x/95/98 started as DOS based).

4] Note on Android/Mac OS.

→ Android:-

- Type: Mobile operating system based on a modified Linux Kernel.
- Environment: Designed for touch devices, apps distributed through app stores, heavy use of Java/Kotlin (Android SDK).
- Features: Multitasking, notifications, app sandboxing, permission needed model, wide hardware diversity.
- Use-cases: Smartphones, tablets, some TVs and wearables.
- Strengths: Large app ecosystem, high hardware variety, customizable.
- Weaknesses: Fragmentation across versions and devices can complicate updates and security.

→ macOS:-

- Type: Desktop operating system by Apple, Unix-based (derived from BSD and Mach kernel).
- Environment: Polished GUI (Aqua), strong ecosystem with Apple hardware (MacBooks, iMacs).
- Features: Integrated UI, security feature (System Integrity Protection), native support for ~~dev~~ development tools (Xcode), robust GUI Apps.
- Use-cases: Creative work, software development, general desktop computing.
- Strengths: Cohesive hardware-software integration, polished UX, good developer tools.
- Weaknesses: Limited hardware choices and higher cost relative to many PCs.

5 | Types of latest processors.

→ A processor (CPU - Central Processing Unit) is the Brain of a computer or mobile device. Modern processors combine high speed, energy efficiency, and integrated graphics with advances in semiconductor technology (down to 3nm & below). Today's processors offer multiple cores, simultaneous multithreading, AI acceleration, and powerful integrated GPUs.

1. Desktop & Laptop processors

a) Intel processors (Core & Xeon families)

• Core 14th Generation (Raptor Lake Refresh, 2024-25)

• Types: Core i3, i5, i7, i9.

• Key features: Hybrid architecture with Performance-cores (P-Cores) and Efficient-cores (E-cores); Intel 7 process node; support for DDR5 RAM and PCIe 5.0; integrated

• Intel UHD/ARC graphics.

• Intel Xeon (Workstation/Server)

• Designed for servers and data centers.

• Large core counts (up to 64+), support for ECC memory, high reliability.

b) AMD Processors (Ryzen & EPYC)

• Ryzen 7000 & 8000 series (Zen 4/Zen 5)

• Built on TSMC 4nm/3nm processes

• Up to 16 cores / 32 threads in mainstream desktop chips.

• PCIe 5.0, DDR5, and advanced integrated Radeon graphics

• Highlight: Strong multi-thread performance and competitive gaming power.

• Threadripper & EPYC (Server/Workstation)

• Massive core counts (up to 96 cores for EPYC Genoa).

• Ideal for heavy computation, virtualization, and cloud infrastructure

2. Mobile & Tablet Processors

a) Apple silicon (for iPhone, iPad, Mac)

• M-Series (M3, M4)

• 3nm fabrication, high performance CPU + GPU + Neural Engine.

• Unified memory architecture for fast data sharing between CPU and GPU.

• Excellent power efficiency for laptops and desktops.

• A-Series (A17 Pro, A18 - iPhone)

• Focus on gaming and AI with powerful GPU cores and dedicated machine-learning accelerators.

b) Qualcomm Snapdragon

• Snapdragon 8 Gen 3 / Gen 4 (Flagship Android)

- Built on 4nm or 3nm nodes
- Octa-core design with performance and efficiency clusters
- Integrated Adreno GPU and Hexagon AI Engine for on-device machine learning

c) MediaTek Dimensity

- Dimensity 9300 / 9400
 - All "big-core" designs for flagship Android phones.
 - Strong AI performance and ~~eff~~ energy efficiency.

d) Samsung Exynos

- Exynos 2400 / 2500
 - Built with Samsung latest 3nm technology.
 - Includes AMD RDNA-based GPU for improved gaming.

Specialized & Emerging Processors

- AI / Neural Processors:
 - Google Tensor G3 (Pixel phones), Apple Neural Engine, and dedicated NPUs in Intel / AMD chips accelerate AI tasks like image recognition and natural language processing.
- ARM-based Laptop chips:
 - Qualcomm Snapdragon X Elite targets Windows laptops with high performance-per-watt and integrated 5G.
- Data Center & High-Performance Computing (HPC):
 - NVIDIA Grace CPU Superchip (ARM-based for servers).
 - Amazon Graviton 3/4 (cloud-optimized ARM processors).